

MOHD KHALID

Software Engineer | Backend & AI

Distributed Systems • AI Systems • Intelligent Automation

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PROFESSIONAL SUMMARY

- Software Engineer specializing in **Node.js**, **TypeScript**, distributed backend systems, AI applications and intelligent automation, serving **10,000+ users** and supporting **100,000+ travel bookings**.
- Founding Backend & AI Engineer experienced in event-driven architectures, AI document intelligence and multi-agent systems, with end-to-end ownership from architecture to production support.

PROFESSIONAL EXPERIENCE

Founding Backend & AI Engineer, **Flyo.ai**

May 2024 – Present

- Collaborated directly with founders and CTO to design backend architecture, evaluate technical trade-offs and deliver production features across consumer and enterprise travel platforms.
- Engineered scalable backend services and APIs powering AI-enabled travel automation, itinerary management and enterprise travel workflows for **10,000+ users**.
- Designed an event-driven Gmail ingestion pipeline that synchronized historical travel emails before transitioning to incremental synchronization using Gmail Watch and History IDs for reliable, idempotent processing.
- Designed an event-driven document processing pipeline where Gmail Watch triggered AWS Lambda for AI-powered document extraction before publishing normalized travel data to Amazon SQS for asynchronous downstream processing.
- Developed AI-powered document intelligence pipelines using LLMs, OCR and prompt engineering, while fine-tuning GPT models and building automated evaluation pipelines to improve structured extraction accuracy and JSON quality.
- Developed supervisor-based LangGraph multi-agent workflows orchestrating booking, travel history, flight tracking and web check-in agents through modular tool routing.
- Designed secure PostgreSQL view-based data access for AI agents, exposing business-relevant fields through read-only abstractions instead of granting direct access to production tables.
- Engineered autonomous airline web check-in workflows combining browser automation, distributed scheduling, passenger document intelligence and asynchronous boarding-pass retrieval.
- Designed a dual-storage architecture using PostgreSQL for transactional data and MongoDB for AI processing artifacts, enabling traceability, debugging and prompt refinement.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript
Backend	Node.js, Express.js, REST APIs, API Design, Microservices, Webhooks, OAuth 2.0, JWT Authentication, Git
AI Systems	AI Agents, LLM Applications, LangGraph, Prompt Engineering, Fine-tuning, LLM Evaluation, Document Intelligence, Structured Data Extraction, OCR, LlamaParse, Browser Use
Distributed	Event-driven Architecture, AWS Lambda, Amazon SQS, Distributed Scheduling, Browser Automation, Webhook Integrations
Databases	PostgreSQL, MongoDB, Redis
Cloud	AWS EC2, AWS S3, New Relic
Integrations	Gmail API, Google Calendar API, Outlook OAuth, WhatsApp Business API, OneSignal, Variflight, DragonPass, Razorpay, Dodo Payments

KEY ENGINEERING SYSTEMS

Intelligent Travel Automation Platform

AI-powered itinerary intelligence

- Designed a unified travel ingestion pipeline consolidating Gmail, Google Calendar, booking confirmations, passport images and manual flight searches into structured user trips.
- Built AI-driven document intelligence and real-time flight intelligence services using LLMs, OCR, aviation APIs, WhatsApp and push notifications.

Autonomous Flight Check-in Engine

Browser automation

- Designed automated scheduling workflows that initiated airline web check-in based on airline-specific check-in windows without user intervention.
- Engineered browser automation (Browser Use) and asynchronous boarding-pass retrieval pipelines that automatically delivered boarding passes through WhatsApp.

Enterprise Travel Operations Automation Platform

Enterprise AI workflows

- Built AI-powered workflows that classified enterprise travel emails, generated operational summaries and intelligently routed hotel and flight requests.
- Collaborated on building a subscription-based travel platform enabling agencies to automate operations, manage bookings, create SEO-friendly websites and streamline subscription management.

EDUCATION

Masai School

2023 – 2024

Full Stack Web Development Bootcamp

Jamia Millia Islamia

2018 – 2022

Bachelor of Technology, Electronics & Communication Engineering